Yurina Nakazato

Curriculum Vitae (last updated July 2025)

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Research Interests

Galaxy formation and evolution, cosmology, first stars and galaxies, cosmological simulations, multi-wavelength observations

Education

Apr. 2023 - Sep. 2025	**Ph.D. student, The University of Tokyo**, Japan Thesis: Formation and Evolution of High-Redshift Galaxies Using Cosmological Simulations and Multi-Wavelength Observations Advisor: Naoki Yoshida
Apr. 2021 - Mar. 2023	M.S. in Physics, The University of Tokyo, Japan Thesis: *Formation and evolution of star clusters and galaxies in the early Universe* Advisor: Naoki Yoshida President's Award (1/3000), Dean's Award (1/128)
Apr. 2017 - Mar. 2021	B.S. in Physics, The University of Tokyo, Japan

Fellowships and Awards

Oct. 2025	The L'Oréal-UNESCO For Women in Science Japan Fellowship
Aug. 2025	The Excellence Award for Final Exam
Feb. 2024	Outstanding Poster Award at the International Symposium on Quantum Electronics 2024
Apr. 2023- present	JSR fellowship
Apr. 2023- present	Japan Society for Promotion of Science (JSPS) Research Fellow, DC1
Mar. 2023	The School of Science Encouragement Award
Mar. 2023	The University of Tokyo President's Award
Mar. 2023	The Excellence Award for Qualifying Exam
Oct. 2021- present	The International Graduate Program for Excellence in Earth-Space Science (IGPEES), the University of Tokyo
Aug. 2021	Best Oral Presentation Award at 51th astronomical meeting for junior researchers

Leadership & Outreach

Jun. 2023	Organizer of Get-Together Event for Women in STEM, the University of Tokyo
Jul. 2022	Public Talk at Women in STEM, School of Science, the University of Tokyo
Dec. 2019	Rikejo Initiative, the Univerisity of Tokyo
Oct. 2019	Invited talk at Dow Chemical Company, Tokyo, Japan

Grants

Apr. 2023-Mar.2026	Evolution of first galaxies with multi-wavelength observations and numerical simulations, 4.2M JPY (30K USD), JSPS Grant-in-Aid for Early-Career Scientists, No. 23KJ0728
Apr. 2023-Mar.2026	Grand Aid from JSR fellowship, 3M JPY (21K USD)

Talks at Conferences and Workshops (Selected)

total: 25 (24 oral talks, 9 peer-reviewed talks)

Sep. 2024 **ILR workshop 2024 @Osaka: From Galaxy Formation to Supermassive Black Holes**, Osaka, Japan

Aug. 2024 **IAUS391**, Cape Town, South Africa

Jul. 2024 **European Astronomical Society (EAS) Annual meeting 2024**, Padova, Italy

Feb. 2024 **The International Symposium on Quantum Electronics 2024**, poster, The University of Tokyo, Japan

Nov. 2023 **Resolving the Extragalactic Universe with ALMA & JWST**, Waseda Univ., Japan

Jul. 2023 **Shedding new light on the first billion years of the Universe**, Marseille, France

Sep. 2022 **The 9th East Asian Numerical Astrophysics Meeting (EANAM9)**, Okinawa, Japan

Aug. 2022 **Star Formation in Different Environments (SFDE) 2022**, Quy Nhon, Vietnam

Jul. 2022 **National Astronomy Meeting (NAM) 2022**, online, hosted by the University of Warwick

Nov. 2021 **IAU Symposium 362: The predictive power of computational astrophysics as a discovery tool**, online

Aug. 2021 **RESCEU Summer School 2021**, online, hosted by the University of Tokyo

Seminars (Selected)

Dec. 2024 **One-day workshop on high-z cosmological simulations**, The University of Tokyo

Jul. 2024 **Special seminar**, JSR corporation, Online

Apr. 2024 **Special seminar**, Scuola Normale Superiore, Italy

Dec. 2024 **Special seminar**, Universidad Autónoma de Madrid , Spain

Dec. 2024 **Special seminar**, Centro de Astrobiología, (CAB, CSIC-INTA), Spain

Aug. 2023 **One-day workshop on galaxies in the era of JWST/ALMA**, The University of Tokyo

Mar. 2023 **Supersonic Project: Collaboration meeting**, UCLA, US

Feb. 2023 **Special seminar**, UCLA, US

Coverage in Media

Hunting Faint Galaxies with JWST as a Test for the Dark Matter Model based on Williams et al. 2024

- *Bright galaxies put dark matter to the test*, UCLA

The core of the most distant galaxy protocluster observed by JWST & ALMA based on Hashimoto et al. 2023

- *The Strong Tag Team of the James Webb Space Telescope and ALMA Captures the Core of the Most Distant Galaxy Protocluster*, Kavli IPMU, Tsukuba Univ., Waseda Univ., Nagoya Univ.

Teaching & Professional Survice

Teaching

Apr. 2021-Aug. 2021	Teaching Assistant of Fluid Mechanics, the University of Tokyo
Apr. 2019- Jan. 2024	Language Assistant, International Lounge, the University of Tokyo

Professional Survice

Apr. 2025	Organizer, One-day workshop on for stellar-cluster simulation, The University of Tokyo, Tokyo, Japan
Dec. 2024	Organizer, One-day workshop on high-z cosmological simulations, The University of Tokyo, Tokyo, Japan
Aug. 2023	Organizer, The 53rd Summer School on Astronomy and Astrophysics, The University of Tokyo, Tokyo, Japan
Mar. 2023	Workshop Organizer, Astrophysics Workshop for Young Researchers, The University of Tokyo, Tokyo, Japan

Proposals

co-Investigator on ALMA proposals

4. 2023.1.00843.S, PI: Y.Tamura, 15.6h
   Disentangling the ionization structure of HII regions in a density-bounded Lya emitter in the Reionization Era
3. 2023.1.00193.S, PI: Y. Fudamoto, 17.3h
   [CII] 158um emission line and dust observation of the most distant known overdensity of galaxies at $z = 7.88$
2. 2023.1.00235.S, PI: Y. Fudamoto, 10.0 h
   The ALMA-JWST synergy: [OIII]88um and [CII]158um emission line observations of $z = 9.51$ *galaxy identified by JWST*
1. 2023.1.01362.S, PI: Y. Tamura, 25.9 h
   Pandora's ELPIS: The Emission-Line Protocluster Imaging Survey of the furthest overdensity beyond Pandora's Cluster

Research & industry experience

Apr. 2024 - Jul. 2024	**Visiting Student at Scuola Normale Superiore di Pisa** - Host researcher is Prof. Andrea Ferrara.
Nov. 2023 - Dec. 2023	**Visiting Student at Universidad Autónoma de Madrid** - Host researcher is Prof. Daniel Ceverino. - Worked on formation and evolution of clumpy galaxies by using FirstLight simulation
Feb. 2023 - Mar. 2023	**Visiting Student at UCLA** - Host researcher is Prof. Smadar Naoz. - Worked on Supersonic Project [link]
Jun. 2020 - Mar. 2021	**Study and Visit Abroad Program** - Funded by the faculty of science, the University of Tokyo - Online research internship at Naoz lab, UCLA
Dec. 2019 - Jan. 2020	**Online Language Exchange Program - Utokyo & TUM-** - Participated in an online international exchange program with students from Technical University of Munich
Jun. 2019 - Sep. 2019	**UTokyo Global Internship Program**

	- Funded by DAIKIN, a company leading air conditioning and refrigeration. - Two-month workplace training at DAIKIN Japan and two-week research internship at DAIKIN Europe in Belgium. Business proposal for food loss and integrated solution of air conditioning and refrigeration.
Aug. 2019	**Summer School of Particle Physics and Nuclear Physics** - Funded by the High Energy Accelerator Research Organization. - Experiment of measuring muon decay time and observing Lamor Precession and Single-Spin Asymmetries. Made the final presentation and poster session.
Aug. 2019	**Nanotechnology Platform Student Training Program** - Funded by National Institute for Materials Science. - Five-day research program at Spring-8, the world's largest third-generation synchrotron radiation facility. Conducted X-ray photoelectron spectroscopy (XPS) experiments and data analysis. Made a presentation at the University of Tokyo in September.
Feb. 2019- Mar. 2019	**Undergraduate Research Assistant, TOMODACHI STEM Program** - Funded by U.S.-Japan Council. - Five-week science & engineering research internship at Gerts lab, Rice University, Houston. Researched and analyzed the heavy iron collision data of STAR experiment conducted at BNL. - Final week study tour to Washington, DC including site visits to the Society for the Promotion of Science, JAXA, U.S.-Japan Council, and Women in STEM Workshop at Leihigh University.
Jul. 2014- Aug. 2014	**Okinawa Global Leaders Program** - Funded by Okinawa Prefectural Board of Education. - Three-week program to introduce students to key concepts in intercultural communication and global leadership.

Active collaborations

- Supersonic Project
- FirstLight simulations
- Reionization and the ISM/Stellar Origins with JWST and ALMA (RIOJA, JWST GO1840)

Competencies

Scientific programming	: Python, C++, Fortran, Latex
Simulation software	: SKIRT, AREPO, ART
Astrophysics tool	: CLOUDY, SKIRT, Astrodendro, A-SLOTH

References

Prof. Naoki Yoshida (current supervisor), The University of Tokyo, naoki.yoshida@ipmu.jp
Prof. Andrea Ferrara, Scuola Normale Superiore, andrea.ferrara@sns.it
Prof. Smadar Naoz, University of California, Los Angeles, snaoz@astro.ucla.edu